

Probabilistic Forecasting of S&P 500 Stocks Using Conditioned Normalizing Flows

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Abstract

Forecasting the behavior of financial markets is a long-standing challenge in computer science, financial data science. Traditional approaches to this problem in industry are to use ARIMA or linear factor models, which often fail to capture the nonlinearity and distributional complexity of financial returns. These oversimplifications, then, don't give us an accurate picture of the risk. Even modern deep learning models, while effective at capturing temporal dependencies, tend to rely on fixed output distributions, like the Gaussian, which limits their ability to represent uncertainty in multi-modal or heavy tailed markets [1].

This thesis investigates the application of Conditioned Normalizing Flows (CNFs) for probabilistic forecasting of daily S&P 500 stock returns. CNFs are generative models that map latent variables drawn from a, usually simple, base distribution (e.g., $\mathcal{N}(0, I)$) into predictive distributions, conditioned on exogenous features. We implement a CNF architecture that combines a recurrent LSTM encoder with RealNVP-based flow transformations, trained within the PyTorchTS time series forecasting framework. The model conditions on lagged returns, calendar-based Fourier features, and macroeconomic indicators such as VIX and IRX, allowing it to identify and adapt to different market regimes.

The primary research question addressed is whether CNFs can offer better equity return forecasts more accurate uncertainty estimates than standard industry models. To answer this, we construct a reproducible dataset of daily log returns and macro indicators from 2019 to 2024, and evaluate the model using probabilistic metrics including the Continuous Ranked Probability Score (CRPS) and interval coverage.

Empirical results demonstrate that CNFs outperform baselines in CRPS, but slightly underperform in interval calibration, indicating some overconfidence in uncertainty estimates. Despite this tradeoff, CNFs provide more expressive and meaningful distributions than traditional models. This work highlights the potential of CNFs as robust, data-driven models for uncertainty-aware forecasting in modern financial markets.

1 Introduction

1.1 Context

Forecasting the behavior of financial markets remains a long-standing problem in statistics, machine learning, and economics. Predictions of future asset returns inform decisions across a range of financial activities including but not limited to long-short strategies, hedging, and risk management. The importance is obvious, yet the task is inherently challenging: asset prices are noisy and are influenced by political regime shifts, policy changes, natural disasters and various other exogenous events which invalidate many modeling assumptions.

Most classical models in time series analysis, such as ARIMA, make a lot of unrealistic assumptions about the linearity and distribution of the underlying data that doesn't hold true for a lot of public markets. More recent deep learning attempts have introduced models like LSTMs and Transformers that can learn temporal dependencies more effectively. However, these models often treat uncertainty modeling as an afterthought, defaulting to Gaussian likelihoods which assume symmetric, unimodal returns. This simplification is especially problematic in financial settings, where return distributions are often heavy-tailed, and regime-dependent making accurate uncertainty estimations crucial to informing risk-based decision making.

To address these limitations, we propose a generative probabilistic framework based on Conditioned Normalizing Flows (CNFs). CNFs modify traditional normalizing flows by incorporating conditioning variables—such as macroeconomic indicators and calendar features—into the density estimation process. This conditioning enables the model to adapt the shape of the output distribution in response to structural signals in the data allowing for more accurate representations of tail risk and market dynamics.

1.2 Research Questions

- Can CNFs provide probabilistic forecasts of asset returns that outperform industry standard models? Can the performance difference be quantified?
- Do macroeconomic indicators such as the VIX and 13-week Treasury yield (IRX) enhance forecasting by capturing exogenous regime information?

1.3 Contributions

The focus of this work is on forecasting daily log returns of S&P 500 stocks using a CNF-based model trained on lagged returns and exogenous macro features. The model is implemented within the PyTorchTS time series framework, based on previous methods using RealNVP flows [1]. This enables rigorous backtesting and evaluation against established baselines. To that end, this report makes the following contributions:

- Implements a CNF model based on a RealNVP architecture with an LSTM encoder.
- Integrates macroeconomic variables (VIX, IRX) and Fourier time embeddings into the conditioning input.
- Constructs a daily-resolution S&P 500 dataset with aligned macroeconomic indicators and Fourier features, processed using a reproducible pipeline.
- Evaluates the model against traditional and deep learning baselines using probabilistic metrics including CRPS and visual forecast plots.

2 Background

2.1 Limitations of Classical Forecasting Models

Traditional point forecasting methods like Autoregressive Integrated Moving Average (ARIMA), have historically been very popular in industry. These models, however, rest on strong assumptions of linearity, and homoscedasticity. In the context of financial time series—particularly equity returns—these assumptions break down frequently. Empirical return distributions are typically heavy-tailed and exhibit volatility clustering, structural breaks, and non-stationary behavior. As a result, point forecasts from ARIMA models, and many other traditional linear models, often fail to account for the true uncertainty in return dynamics [2].

Multivariate extensions such as Vector Autoregressive models attempt to capture inter-series dependencies but suffer from high parameter complexity, especially when modeling hundreds of assets, such as in the S&P 500 universe [3]. They also struggle to incorporate exogenous features like macroeconomic indicators, further limiting their practical forecasting power [4].

2.2 Deep Learning Models for Time Series

Neural network-based models have demonstrated strong empirical performance across domains involving time-series, or sequential data [5]. In financial applications, LSTMs are an improvement over ARIMA because they can capture nonlinear temporal dependencies and have longer-range memory effects. However, despite their flexibility in feature learning, standard LSTM forecasting models generally assume a fixed output distribution, most commonly Gaussian. This simplification results in underestimation of tail risk and limited ability to capture the heteroscedastic, asymmetric distributions commonly found in public markets.

Models like DeepAR and DeepVAR extended LSTM architectures to support probabilistic forecasts over multiple time series. Yet even these models rely on parameterizing likelihoods with fixed distributions and do not directly learn the full return distribution from data [6].

2.3 Normalizing Flows

Normalizing Flows (NFs) are a method for constructing complex distributions by transforming a latent probability density into our target through a series of invertible mappings.

That is, given a bijective function f and a base density $p_Z(z)$, then the target variable $X = f^{-1}(Z)$ follows the distribution by the change-of-variable formula:

$$p_X(x) = p_Z(f(x)) \left| \det \frac{df(x)}{dx} \right|^{-1}.$$

This formula lends itself to exact computation of log-likelihoods. For any sample x we can calculate it easily, and exactly using the formula. Unlike other deep learning models like Variational Autoencoders or GANs flows can be trained directly by maximizing the exact log-likelihood of observed data. In architectures like RealNVP, the structure of the transformation f is designed to make the Jacobian determinant both efficient and tractable to compute. This allows the model to scale to high-dimensional outputs while still preserving the ability to learn asymmetric, non-Gaussian distributions, solving our initial issue.

2.4 Conditioned Normalizing Flows

NFs offer flexibility and improvements in modeling distributions, but aren't inherently suited to complex time-series forecasting. A key limitation is that standard NFs treat samples independently and do not incorporate temporal or contextual information; CNFs address this by conditioning the transformation f on contextual variables C_t :

$$Z_t = f(X_t | C_t).$$

This conditioning enables the model to adapt the transformation based on covariates such as lagged features, calendar embeddings, identity embeddings, or macroeconomic signals. The result is a flow that evolves with market dynamics, learning a time-varying distribution over returns. In our specific use case, we condition on macroeconomic variables in an attempt to capture broader patterns and relationships.

We build upon the TempFlow framework proposed in [1], which uses a recurrent encoder to aggregate conditioning information and applies RealNVP flows to model the target distribution. This allows the model to capture both intertemporal dependencies and nonlinear effects of exogenous signals such as the VIX and IRX.

3 Methodology

3.1 Overview

This section outlines the modeling approach used in this report, which is based on the architecture proposed in [1]. The model consists of a recurrent encoder, temporal and macroeconomic feature processing, and a RealNVP-based flow decoder that defines the target distribution.

3.2 Data and Preprocessing

We construct a dataset of S&P 500 data spanning the period from January 2019 to January 2024. The raw data is sourced from Yahoo Finance using the `yfinance` API and consists of:

- Equity Price Data: Daily open, close, high, and low prices for all S&P 500 constituents, filtered for continuity and liquidity. The relevant portion for our purposes is the series of close prices.
- Macroeconomic indicators: Daily values of the CBOE Volatility Index (VIX) and the 13-week Treasury bill yield (IRX). VIX representing volatility of the market, and widely known to be inversely correlated with S&P 500 returns should be a good indicator of risk in the market. IRX is commonly accepted as risk-free rate in the US economy, and is a good benchmark for investor sentiment.

For each equity series, $P = (P_1, \dots, P_n)$, we compute log returns by the formula: $r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$, align all series on a common calendar, and ensure all series are standardized at the asset level. Fourier time features are added to represent seasonal patterns. Essentially they transform the input date data to ensure our model captures week-day/weekend/holiday dynamics.

3.3 Forecasting Task

Let X_t be a time series of daily log returns for a given stock, and let C_t be the set of conditioning features available at each time step t , which includes lagged values of X_{t-k} , our Fourier time features, and macroeconomic indicators. The objective is to learn the conditional distribution of the next h steps:

$$p(X_{t+1}, \dots, X_{t+h} \mid X_t, \dots, X_{t-\ell+1}, C_{t+1}, \dots, C_{t+h}),$$

where ℓ is the context length and h is the prediction window length. How we accomplish this is described below.

3.4 Architecture

The CNF architecture consists of two primary components and is detailed in this section and outlined in Figure 3.1:

- **Reccurent Encoder:** A two-layer LSTM processes lagged inputs and conditioning features to generate a hidden representation $h_t \in \mathbb{R}^d$ for each time step.
- **RealNVP Flow:** A RealNVP transformation conditions on h_t to define a bijective mapping $f_\theta(\cdot \mid h_t)$ between the latent Gaussian space and the target return distribution.

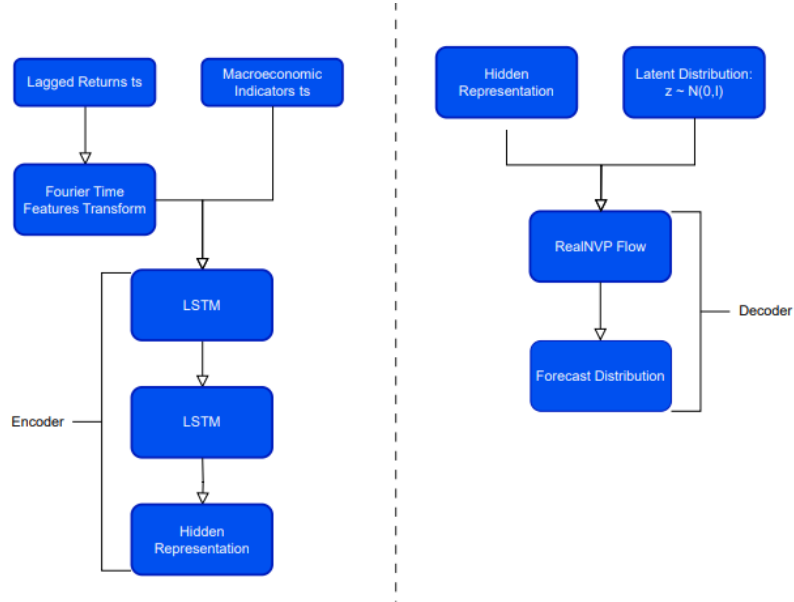


Figure 3.1: Overview of the CNF forecasting architecture. An LSTM encoder processes lagged returns, Fourier time features, and macroeconomic indicators to produce a hidden representation h_t . A RealNVP flow conditions on h_t and transforms a latent sample $z \sim \mathcal{N}(0, I)$ to produce a forecast distribution.

3.4.1 Conditioning Features and Encoding

The input to the encoder at each time step t is a concatenation of four feature types:

- **Lagged returns:** $[X_{t-1}, X_{t-2}, \dots, X_{t-10}]$.
- **Fourier time embeddings:** Sinusoidal encodings to model week/month/seasonality in our calendar.

- Macroeconomic indicators: VIX and IRX, capturing forward-looking volatility and monetary policy signals.
- Series identity embeddings.

These feature vectors are concatenated and passed through a two-layer LSTM. The LSTM outputs a contextual hidden representation h_t that summarizes relevant past behavior and external conditions which is then passed into our decoder.

3.4.2 RealNVP Flow Decoder

The decoder consists of multiple RealNVP blocks, each applying a coupling transformation to a latent input $z \sim \mathcal{N}(0, I)$ conditioned on h_t . Each block transforms half of the input using the other half, through an affine transformation:

$$\begin{aligned} x_1 &= x_1, \\ x_2 &= x_2 \cdot \exp(s(x_1, h_t)) + t(x_1, h_t), \end{aligned}$$

Here, s and t are neural networks conditioned on the LSTM output h_t . Stacking multiple blocks yields a deep transformation while preserving invertibility. The structure of RealNVP ensures that the Jacobian of the transformation is triangular, enabling efficient computation of its log-determinant. This property allows for exact likelihood evaluation during training.

3.5 Training

To train the model, we use maximum likelihood estimation (MLE). For each observed return X_i and its associated context C_i the encoder produces a hidden representation h_t . This representation conditions the RealNVP flow, which transforms X_i into a latent variable $z = f_\theta(X_i | h_t)$. The model then evaluates the log-likelihood of z under the standard Gaussian prior and adjusts for the volume change of the transformation using the Jacobian determinant.

The resulting objective minimizes the negative log-likelihood:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p_Z(f_\theta(X_i | C_i)) + \log \left| \det \frac{\partial f_\theta}{\partial X_i} \right|.$$

3.6 CRPS Evaluation

In this report, one metric we use to evaluate our output densities is the *Continuous Ranked Probability Score* (CRPS). The CRPS in essence it generalizes mean absolute error to distribution predictions. Precisely, for a forecast CDF F and ground truth x :

$$\text{CRPS}(F, x) = \int_{-\infty}^{\infty} (F(z) - \mathbf{1}\{x \leq z\})^2 dz.$$

Where 1 is the Heaviside step function. Lower CRPS generally indicates a forecast that is both sharp and well-calibrated. Additional metrics used include empirical coverage of predictive intervals and visual inspection of forecast trajectories compared to realized outcomes.

4 Results

4.1 Quantitative Evaluation

Table 4.1 summarizes the CRPS performance of the CNF model compared to the Gaussian likelihood baseline across 100 randomly selected forecast windows. Each result is computed as the mean over five trained models and 20 test windows each.

Model	Mean CRPS	Standard Deviation
CNF (ours)	0.0138	0.0112
Gaussian LSTM	0.0153	0.0172
ARIMA	0.0165	0.0194

Table 4.1: Average CRPS over 100 forecast windows (5 models $\times$ 20 samples). Lower is better.

4.2 Interval Calibration

We also assess the empirical coverage of the CNF’s 80% predictive interval by evaluating how often the ground-truth value falls between the 10th and 90th percentiles of the forecasted distribution. Table 4.2 reports the average coverage over the same 100 test windows.

Model	Nominal Coverage	Empirical Coverage
CNF (ours)	80%	71.59%
Gaussian LSTM	80%	86.88%
ARIMA	80%	87.44%

Table 4.2: Empirical coverage of 80% prediction intervals.

4.3 Qualitative Examples

For each forecast window, we generate plots displaying historical training data (black line), predictive mean and interval (dashed black line with 80% error interval shown), and ground truth holdout values (blue markers). Various examples are shown in the figures below:

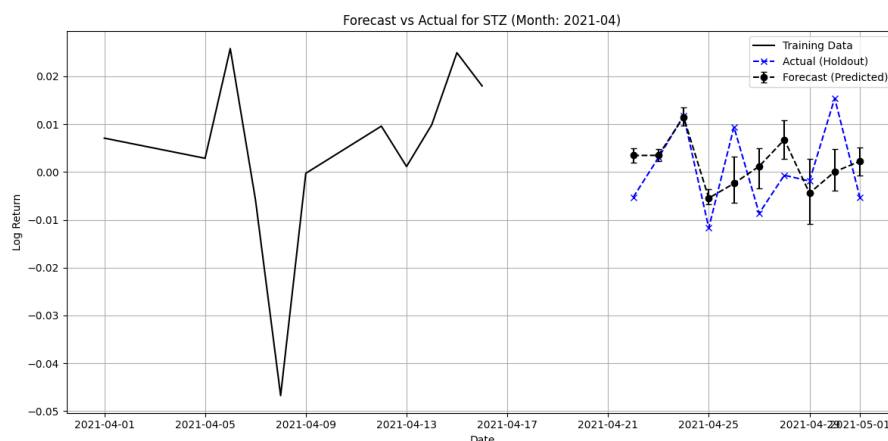


Figure 4.1: Forecast output of our CNF model on Constellation Brands (STZ) stock for Apr 2021.

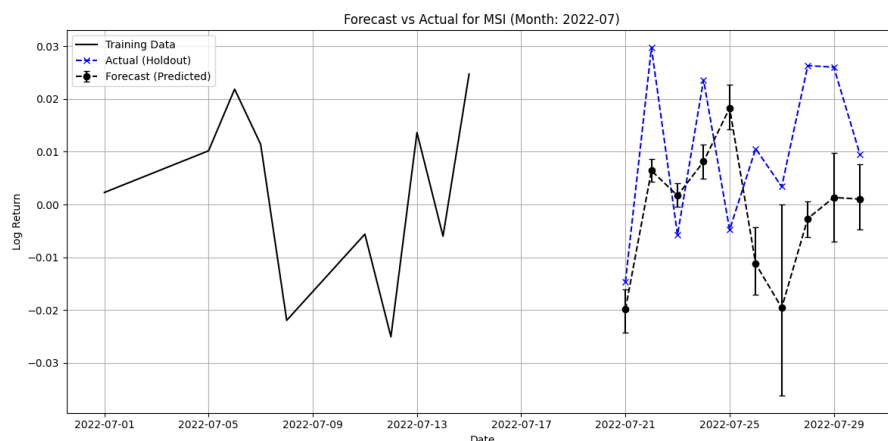


Figure 4.2: Forecast output of our CNF model on Motorola Solutions Inc (MSI) stock for Jul 2022.

Qualitative assessment of the plots above, and numerous other plots generated, show the CNF's predictive intervals adjust adaptively to local volatility, widening during macro-driven shocks such as the widening of error bars in 2021 to account for COVID-19 and in

2022 for the war in Ukraine. Furthermore, forecast distributions are often visibly skewed, reflecting asymmetry in expected returns.

4.4 Interpretation of Results

The CNF outperforms the Gaussian LSTM and ARIMA in both mean score and standard deviation, indicating slightly better calibration of its forecast distributions. While all three models exhibit relatively low CRPS values due to the short horizon and log scaling, the difference is consistent across multiple seeds and assets. The improvement in CRPS, although modest, is meaningful for applications requiring fine-grained calibration, such as options pricing. Furthermore, in the case of many large financial firms, small percentage differences result in shifts of millions of dollars in the economy only giving more credence to a small improvement being significant.

The CNF's empirical coverage for its 80% predictive interval falls short, indicating under-coverage and thus slightly overconfident intervals. In contrast, the Gaussian LSTM and ARIMA models both exhibit over-coverage. This suggests that although those models produce wider, more conservative prediction intervals, they may be less precise in quantifying uncertainty, which was our goal in using a CNF. The model's reliance on macroeconomic signals may introduce noise or overfitting in calm periods which may partially explain this.

5 Conclusions and Future Work

This thesis explored whether Conditioned Normalizing Flows (CNFs) are effective in forecasting of daily S&P 500 stock returns, and if they outperform traditional models. The central research question asked whether flow-based generative models could better capture the distributional complexity of financial returns and deliver more calibrated uncertainty estimates.

Empirical results offer a mixed perspective. On one hand, the CNF model achieves the lowest mean CRPS across 100 forecast windows, outperforming both Gaussian LSTM and ARIMA baselines trained on the same data and seeds. This indicates that CNFs can produce sharper and more accurate probabilistic forecasts in terms of average prediction error. On the other hand, the CNF underperforms both baselines in terms of interval calibration, suggesting that while CNFs generate tighter distributions, they are more prone to underestimating uncertainty—missing the true value more often than their more conservative counterparts.

Despite this tradeoff, CNFs demonstrate clear modeling advantages. Their ability to represent skewed and multi-modal return distributions enables them to adapt more responsively to changing market regimes. For instance, the model produces wider forecast intervals during periods of elevated VIX. This falls, in line with a known macroeconomic concept called volatility clustering, giving confidence that the model is learning real conditioned patterns.

Nonetheless, several limitations and potential improvements remain. The current implementation models assets independently and does not capture cross-asset or sectoral dependencies. The conditioning on macroeconomic indicators implicitly helps this a little with how VIX and IRX respond to the market, but much more could be done. Incorporating cross-series attention mechanisms could allow the model to dynamically learn dependencies across assets and sectors, potentially improving multivariate forecasting performance. Also, RealNVP-based flows impose significant computational overhead during training. Flow matching is simulation free, and when compared to traditional CNF, it should be more efficient and scalable. Furthermore, guided flow matching, which applies methods from reinforcement learning to impose more principled learning schemes using a reward function, could lead to more sensible predictions.

Despite these challenges, CNFs represent a promising direction for modern financial forecasting. By learning distributions directly—rather than assuming fixed forms—CNFs enable better risk-aware modeling and open doors to applications like real-time value at risk estimation, more volatility-conscious trading strategies, and enhanced stress testing pipelines.

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